

SECTION 15050-A BASIC MECHANICAL REQUIREMENTS

PART 1 -- GENERAL

1.1 WORK INCLUDED

- A. This Section specifies the basic requirements for all sections included in Division 15 and sections in Divisions 11 and 13 that specifically reference this section.
- B. Requirements for the following are included:
 - 1. Related work (other Contract Documents and specification sections) which must be combined with the requirements of this Section.
 - 2. Design criteria.
 - 3. Regulatory requirements.
 - 4. Delivery, storage, and handling.
 - 5. Submittals.
 - 6. Product quality, basic type, and finishes.
 - 7. Equipment identification.
 - 8. Vendor representative requirements.
 - 9. Mounting and shimming.
 - 10. Inspection.
 - 11. Safety considerations.
 - 12. Cleaning, startup, and adjustments.

1.2 RELATED WORK

- A. Use this general section in conjunction with the following other specifications and related Contract Documents to establish the total requirements for the project equipment and systems.
- B. CAUTION: Use of this Section without including all of the above-listed items will result in omission of basic requirements.

1.3 DESIGN CRITERIA

A. Supports, Anchors, and Restraints:

1. General:
 - a. Contractor is responsible for design of supports, anchors, and seismic restraints that are not shown on Drawings; including such requirements for equipment, conduit, piping, and ductwork.
 - b. Seismic restraints and anchors shall resist seismic forces as specified for the project location.
 - c. Seismic restraints shall not introduce stresses in piping or ductwork caused by expansion or contraction due to changes in temperature.
 - d. Connections to structural framing shall not deform the framing members. Provide supplementary steel as required.
 2. Suspended Piping and Ductwork Seismic: restraints shall be in accordance with the latest edition of the SMACNA Seismic Restraint Manual - Guidelines for Mechanical Systems for the SHL of the project location.
 3. Engineered Support Systems: The following support systems shall be designed, detailed, and bear the seal of a professional engineer registered in the area of jurisdiction:
 - a. Supports and seismic restraints for suspended equipment.
 - b. Support frames, such as pipe racks or stanchions, which provide support for piping, ductwork, and equipment from below.
 - c. Equipment and support frame anchors supported from slab or structure.
- B. Compliance by the Contractor with the provisions of this Specification does not relieve them of the responsibilities of furnishing equipment and materials of proper design, mechanically suited to meet operating guarantees at the specified service conditions.
- C. The flange facing for nozzles, manways, handholes, and equipment body joints shall be correct for the type of

gasket provided. Ordinary compressed sheet gaskets require only the standard machine finish on the raised face. Smooth metal gaskets, either plain or filled, require a fine-serrated finish. Corrugated metal and spirally wound metal gaskets, either plain or filled, require a smooth finish (125 AARH or 125 RMS) for proper applications. Ring-type joint gaskets and gasket grooves shall conform to ANSI B16.20.

1.4 REGULATORY REQUIREMENTS

- A. Obtain permits and arrange inspections required by codes applicable to this Section. Submit written evidence to the Engineer that required permits, inspections, and code requirements have been secured.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Preparation for Shipment:

1. Prepare equipment for the specified shipment. Ship each unit or part of unit for outdoor storage for a period of at least 6 months in a manner requiring no disassembly prior to operation.
2. Adequacy of the preparation for shipment provisions employed with respect to materials and application is the sole responsibility of the Contractor.
3. Protect flange facings by securely fastened durable covers to prevent damage during shipment.
4. Provide equipment completely free of water prior to shipment preparation.
5. Apply a heavy coating of rust-inhibiting compound to exposed finished and machined surfaces, including bolting. Spray or coat internal metal surfaces with a suitable rust preventative which can be easily removed.
6. Bearings, bearing housings, and oil systems; including reservoirs, coolers, filters, and piping; shall be thoroughly cleaned internally and coated with a suitable rust preventative prior to shipment.
7. Provide adequate protection against mechanical damage and atmospheric corrosion in transit.
8. Mount and/or secure instruments and valves, including auxiliary systems, to eliminate damage during shipment, storage, operation, and maintenance.

9. Provide supports and rigging connections to prevent damage during shipment or delivery.
10. Provide threaded ends with a pipe plug of the same material as the connections using Teflon tape thread sealant.
11. Suitably mark openings that have applied rust preventive protection to indicate the type and name of the rust preventive material.
12. Suitably mark equipment at openings to indicate the nature of insulating oils or other fluids contained within including shipping and storage precautions.
13. Cap open ends of tubes for protection, taping without secure placement of a cap is not adequate for this purpose.
14. Pack, securely anchor, and weather-protect equipment for the shipment method required. Box separate, loose, or spare parts with each part individually protected as required. Mark each individual container both inside and outside with the equipment number and service for which the parts are intended.
15. Pack and ship with the equipment one complete set of installation, operating, and maintenance instructions that are in addition to the sets to be sent directly to Owner.
16. Include in each container a complete bill of materials identifying each part and mark each part so it can be compared with the bill of materials.
17. Impression stamp piping connections furnished on the purchased equipment to agree with manufacturer's connection table listed on the general arrangement drawing. Marking instead of stamping is only acceptable where the connections cannot be impression stamped due to size or configuration.

1.6 SUBMITTALS

A. Provide the following:

1. With the Bid:

- a. An individual price and delivery schedule for each equipment item.

- b. The length of time required for certification of all information and documentation.
- c. Estimated equipment weight and support requirements.
- d. Electrical power requirements, including in-rush current and recommended fuse size.
- e. Utility requirements.
- f. A list of deviations from the specified requirements.
- g. Adequate dimensional data to permit the design of foundation, piping, and wiring connections.
- h. Clearance required for disassembly for maintenance or service.

2. Within 4 weeks of award of Contract:

- a. Certified correct dimensional drawings of completed assembled units, which show:
 - (1) Identification data for equipment components.
 - (2) Rotation.
 - (3) Weight.
 - (4) Location of motor junction box(es).
 - (5) Piping connections identified with the size, rating, and facing indicated.
- b. Identify auxiliary connections on the composite outline drawing as follows:
 - (1) Not furnished this order.
 - (2) Not drilled this order.
 - (3) Plugged, requires field piping by Contractor.
 - (4) Plugged, not required this order.
 - (5) Piped by manufacturer.
- c. Reference to supplementary drawings required to complete auxiliary piping.

- B. Review of manufacturer's drawings shall not relieve the Contractor from the responsibility for equipment performance and compliance with this Specification.

PART 2 - PRODUCTS

2.1 GENERAL

- A. Equipment shall be the manufacturer's most recent standard design. Equipment designs which have not been previously used successfully in an industrial application are not acceptable.
- B. Provide 1 liter and 2 spray cans of each standard paint used from each equipment and/or materials supplier for touchup. Adequately label paint cans to identify where they are to be used. When equipment is involved, the label shall, at a minimum, include the Owner's equipment tag number.

2.2 EQUIPMENT IDENTIFICATION

- A. A stainless steel identification tag shall be permanently affixed to equipment in an easily accessible location with the following minimum information, as applicable:
 - 1. Manufacturer's name.
 - 2. Manufacturer's model number.
 - 3. Date of manufacture.
 - 4. Design operating conditions.
 - 5. Design pressure and temperature.
 - 6. Serial number.
 - 7. Materials of construction.
 - 8. Code stamp.
 - 9. RPM.
 - 10. Electrical power requirements.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Installation shall be in accordance with the requirements of the equipment manufacturer.
- B. Equipment Manufacturer's Responsibility and Services:
 - 1. A manufacturer's representative for major equipment and operating systems shall be provided as necessary to assist the Contractor during installation and to provide written certification that the equipment has been installed as specified and in accordance with the manufacturer's requirements.
 - 2. The manufacturer's representative shall provide the initial startup of equipment in the presence of the Engineer and Owner.
 - a. Provide a prestart check of piping, valves, control devices, control panels, and equipment.
 - b. Calibrate and adjust equipment and controls for operation at the specified design conditions.
 - c. Provide a record of startup events noting problems and their resolution.
 - d. Provide a record of set points for operational controls and devices.
 - 3. Upon the completion of equipment startup, provide instructional time with the Owner's personnel to review the operations and maintenance manuals and perform each step necessary for startup, shutdown, troubleshooting, and routine maintenance. The instructional time shall be scheduled through the Owner.
 - 4. Upon completion of the inspections, startup, testing, and checkout procedures; the equipment manufacturer shall submit written notice to the Owner and the Engineer that the units are ready for use by the Owner.

3.2 MOUNTING AND SHIMMING

- A. Mount equipment as shown on the Drawings. Provisions for mounting special equipment on spring isolators, snubbers, and inertia bases are specified.
- B. Level the equipment by means of steel wedges (steel plates and steel shims). Wedge taper shall not be greater than 1 millimeter per 10 centimeters (1/4 inch per foot). Use

double wedges to provide a level bearing surface for the equipment. Secure each pair of wedges in their final positions with one tack weld on each side after leveling is complete. Wedging shall be done in a manner that will prevent a change in level or springing of the baseplate when the anchor bolts are tightened.

1. Adjust rotating equipment assemblies such that drivers, shafts, couplings, and driven equipment are aligned, plumb, and level.
- C. Rotating equipment shall be checked for proper alignment with dial indicators. The alignment must be within the tolerances required by the equipment manufacturer. The final alignment check shall be witnessed by the Engineer.
- D. After the equipment has been set in position, aligned, and shimmed to the proper elevations, grout the space between the bottom of the baseplate and the concrete foundation and/or inertia base with a poured non-shrinking grout as specified in Contract.

3.3 INSPECTION

- A. Inspect the work to ensure the installation and workmanship are in accordance with Contract and acceptable industry standards for the work.
- B. Testing shall conform to requirements of Contract, applicable codes and standards, acceptable industry practice, and manufacturer's recommendations.
- C. Contractor is responsible for furnishing the inspector with necessary material certifications, shop test data, radiographic plates, and all other pertinent documentation necessary to verify the Contractor's compliance with the specifications for Contractor-furnished equipment.
- D. No surfaces or parts are to be painted until the inspections are complete.
- E. Materials, equipment, and workmanship are subject to inspection at any time by the Owner. Contractor shall correct any work, materials, or equipment not in accordance with the Contract Documents.

3.4 SAFETY CONSIDERATIONS

- A. Install equipment with suitable clearance for maintenance access or removal of replaceable parts and components. Include unions or flanges as required to perform the

maintenance or removal without disturbing any component except the part requiring removal.

- B. Where equipment requiring periodic maintenance cannot be reached by normal walkways because of interference with ductwork, piping, or other obstructions, provide a safe means of access. Such safe means of access shall conform to OSHA standards for safe maintenance procedures.

3.5 CLEANING, STARTUP, AND ADJUSTING

- A. The Contractor shall be responsible for proper operation of systems, subsystems, and services provided under Contract. Coordinate startup procedures, calibration, and system checkout with all subcontractors involved. Problems shall be diagnosed, corrective procedures shall be initiated, and checks shall be conducted to verify the effectiveness of corrections. Unresolved problems shall be brought to the attention of the Engineer and further steps taken until the required performance is achieved.
- B. Thoroughly clean all parts of the installation upon completion of the construction and before the work will be considered complete. Remove from the premises all refuse material, crates, and rubbish from work. Remove, clean, and reinstall all filters. Check all belt-drive tensions and alignments. Lubricate motors and bearings in accordance with the manufacturer's recommendations prior to equipment startup. Provide a lubrication schedule for each item of equipment furnished under contract, including type of lubricant and application frequency.

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